

1. Administrative & Study Identification

Full Study Title: A Phase 1 Study of PF-08046049/SGN-BB228 in Advanced Melanoma and Other Solid Tumors **Short Title/Acronym:** SGNBB228-001 **Protocol Number:** SGNBB228-001 / C5871001 **NCT Number:** NCT05571839 **Sponsor Name:** Seagen, a wholly owned subsidiary of Pfizer **Investigational Product:** PF-08046049 / SGN-BB228 (Antibody-Anticalin fusion bispecific directed to CD228 and 4-1BB) **Phase of Development:** Phase 1 **Medical Monitor:** Pfizer Clinical Trial Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and to determine the optimal dose and schedule of PF-08046049/SGN-BB228 in participants with advanced melanoma and other solid tumors. **Secondary Objectives:** To evaluate the antitumor activity and efficacy of the investigational product, specifically focusing on Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS).

2.2 Background & Rationale To address the critical need for novel therapeutic options in advanced solid tumors that are relapsed, refractory, or intolerant to standard-of-care treatments. CD228 is a tumor-associated antigen selectively expressed by multiple tumor types, and 4-1BB is an inducible costimulatory receptor on immune cells. SGN-BB228 aims to bind to cancerous cells and stimulate the immune system to attack them.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation, dose optimization, and dose expansion) **Blinding:** Open-label **Randomization:** N/A (Sequential Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 275 participants worldwide (45 in Part A, 30 in Part B, ~200 in Part C) **Number of Sites:** Multicenter across the United States, Canada, France, Germany, Switzerland, and the United Kingdom **Estimated Study Start:** 03-Jan-2023 **Estimated Primary Completion:** 06-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years; All Genders. 2. Histologically or cytologically confirmed metastatic or unresectable solid tumors (Part A/B: Cutaneous melanoma; Part C: Cutaneous melanoma, Non-small Cell Lung Cancer (NSCLC), Colorectal Cancer, Pancreatic Cancer, or Mesothelioma). 3. Disease that is relapsed, refractory, or intolerant to standard of care. For cutaneous melanoma, patients must have been previously treated with an anti-PD-1 or anti-PD-L1 agent.

4.2 Exclusion Criteria 1. History of another malignancy within 3 years before the first dose of study drug, or any evidence of residual disease. Exceptions are malignancies with a negligible risk of metastasis or death. 2. Active central nervous system (CNS) metastases or leptomeningeal disease. (Note: Previously treated brain metastases are allowed if clinically stable for ≥ 4 weeks, no new/enlarging lesions, and off corticosteroids for ≥ 7 days prior to first dose). 3. Melanoma subtypes including acral, uveal, and mucosal are excluded. Prior therapies targeting CD228 or 4-1BB are prohibited.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: SGN-BB228 administered via infusion on either a 21-day cycle or a 28-day cycle (twice during the 28-day cycle) to determine the best schedule. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Cycles are repeated until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing associated side effects. **Prohibited:** Immunotherapy, biologics, and/or other approved or investigational antitumor treatments that were not completed 4 weeks prior to the first dose (or within 2 weeks if underlying disease has progressed on treatment).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans, Pre-treatment biopsy or archival tissue submission.
Baseline	Day 1	First IV Infusion, Safety Labs.
Treatment Cycles	Every 21 or 28 Days	Ongoing IV Infusion of SGN-BB228, Safety Assessments. Scans performed every 6 weeks to evaluate tumor response.
End of Study / Follow-Up	30 Days Post-Treatment	Final safety evaluation, recording of Adverse Events. Survival follow-up ongoing for approx. 1 to 2 years.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability evaluated by the number of participants with dose-limiting toxicities (DLTs) and adverse events (AEs). **Measurement Method:** Clinical assessment, graded severity of untoward medical occurrences through 30 days after the last study treatment.

7.2 Secondary & Exploratory Endpoints - Objective Response Rate (ORR) assessed per RECIST v1.1. - Duration of Response (DOR). - Progression-Free Survival (PFS). - Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of any untoward medical occurrences through 30 days post-last treatment (approx. 7 months total time frame). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timelines for Phase 1 oncology trials. **Data Monitoring Committee (DMC):** Internal sponsor safety review team assigned to monitor dose-escalation (Part A) clearing and MTD/RP2D decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety population includes all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Approximately 275 participants appropriately powered to establish safety (Dose Escalation) and observe preliminary antitumor signals (Dose Expansion). **Handling of Missing Data:** ORR and 95% Confidence Intervals (CI) will be estimated using descriptive statistics. DOR, PFS, and OS will be estimated using the Kaplan-Meier method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical monitoring by the Sponsor (Seagen/Pfizer). **Audit Trail:** Secure eCRF systems deployed for tracking dose escalations, scan assessments, and clinical adherence.

11. Study Identifiers & Governance

Primary ID: C5871001 / SGNBB228-001 **Registry Number:** NCT05571839
Sponsor/Lead Organization: Seagen, a wholly owned subsidiary of Pfizer **Study Title:** A Phase 1 Study of PF-08046049/SGN-BB228 in Advanced Melanoma and Other Solid Tumors **Official Regulatory**

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12. Clinical Framework (Solid Tumor Specific)

Primary Condition: Advanced Cutaneous Melanoma, NSCLC, Colorectal Cancer, Pancreatic Cancer, Mesothelioma **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic or unresectable disease **Medical Methodology:** Targeted Immunotherapy (CD228 x 4-1BB costimulatory bispecific antibody) **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose (RP2D) identification

13. Study Procedures & Methodology

Management Pathway: Stepwise dose escalation (Part A), dose and schedule optimization (Part B), and disease-specific dose expansion (Part C) **Education Protocol (HCP):** Protocol adherence and AE/toxicity management training for Phase 1 investigative sites. **Education Protocol (Patient):** Informed consent detailing the risks of Phase 1 experimental treatments, visit expectations, and scan schedules. **Quality Audit Mechanism:** Independent central/site review of RECIST v1.1 radiographic scans.

14. Observation & Follow-Up Schedule

Total Duration: Approximately 1 to 2 years per participant depending on treatment response **Onsite Visit Cadence:** Every 21 or 28 days corresponding to treatment cycles **Radiographic Monitoring Frequency:** Scans conducted every 6 weeks **Checklist Review Interval:** Regular safety reviews preceding each treatment cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				